

KRASNOZERSKOE, Russian Federation

WMO: 298130

Lat: **53.989N**Lon: **79.253E**Elev: **472**StdP: **14.45**Time Zone: **7.00 (E07)**Period: **94-18**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.9	-25.5	-38.1	0.6	-29.8	-31.7	0.9	-25.0	25.9	18.7	23.1	15.5	4.3	210	0.631

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	86.7	65.3	83.5	65.3	80.5	64.1	69.5	79.9	67.8	78.0	66.3	76.6	9.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	98.2	73.4	64.2	91.5	72.4	62.5	86.0	71.0	34.0	79.7	32.5	78.0	31.3	76.7	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	20.4	17.8	DB	-34.3	92.1	6.4	3.5	-38.9	94.6	-42.6	96.6	-46.2	98.5	-50.9	101.1
			WB	-35.1	72.7	6.8	2.5	-40.0	74.5	-44.0	76.0	-47.8	77.4	-52.8	79.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	-0.5	4.1	19.9	40.3	54.5	65.5	67.4	62.8	52.3	38.2	20.8	6.8	
	DBStd	26.26	14.83	12.63	10.97	11.02	9.67	7.33	5.87	6.79	8.89	9.36	12.82	14.41	
	HDD50	6855	1566	1286	934	327	67	2	0	2	75	380	876	1340	
	HDD65	10825	2031	1706	1399	742	344	83	43	122	393	831	1326	1805	
	CDD50	1805	0	0	0	35	207	466	538	400	144	15	0	0	
	CDD65	300	0	0	0	0	20	97	116	55	12	0	0	0	
	CDH74	2828	0	0	0	12	272	873	928	581	161	1	0	0	
	CDH80	754	0	0	0	1	63	239	251	150	50	0	0	0	
Wind	WSAvg	8.1	8.2	8.5	9.2	8.8	9.0	7.4	6.4	6.8	7.3	7.6	8.7	9.1	
Precipitation	PrecAvg	13.7	0.7	0.5	0.6	0.7	1.0	1.7	2.6	1.7	1.2	1.2	1.0	0.8	
	PrecMax	18.3	1.4	1.1	1.2	1.5	2.2	4.0	6.3	4.9	2.7	3.1	2.4	1.9	
	PrecMin	8.5	0.1	0.0	0.1	0.0	0.0	0.4	0.6	0.1	0.1	0.3	0.5	0.2	
	PrecStd	3.0	0.4	0.3	0.3	0.4	0.5	0.9	1.5	1.1	0.7	0.6	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.3	33.0	43.1	75.2	86.2	89.5	90.6	89.3	86.7	67.9	47.6	34.8	
		MCWB	32.5	31.2	36.8	54.9	61.6	66.3	67.8	65.4	61.7	53.0	44.1	32.7	
	2%	DB	28.8	30.1	39.3	68.3	80.7	85.9	86.0	84.1	78.0	61.7	42.5	32.3	
		MCWB	27.6	28.7	35.1	52.4	59.7	64.8	67.1	65.6	58.9	49.8	38.9	31.3	
	5%	DB	24.9	26.8	36.6	63.1	76.6	82.5	82.5	80.1	72.4	56.3	39.0	29.3	
		MCWB	23.8	25.6	33.5	50.1	58.0	64.7	67.0	64.1	56.8	47.4	36.5	28.1	
	10%	DB	21.3	22.9	34.2	57.5	71.6	79.3	79.2	76.6	67.4	51.5	35.0	26.0	
		MCWB	20.3	21.8	31.8	47.0	55.6	63.6	66.0	62.0	54.7	44.7	32.9	24.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.5	31.5	38.2	57.4	65.1	70.8	72.0	71.3	63.9	54.5	44.8	33.1
			MCDB	33.2	32.4	41.8	70.7	80.0	84.6	81.1	80.6	80.4	64.2	46.8	33.9
2%		WB	27.8	28.8	35.6	54.1	62.3	68.1	70.1	67.7	60.5	51.2	39.6	31.5	
		MCDB	28.8	29.9	38.7	66.1	77.4	80.6	79.8	78.3	74.5	60.2	42.1	32.4	
5%		WB	23.9	25.6	33.8	50.8	59.4	66.4	68.6	65.7	57.9	48.1	36.3	28.1	
		MCDB	24.8	26.7	36.3	61.5	72.9	78.4	78.1	77.1	70.2	54.4	38.9	29.1	
10%		WB	20.3	21.9	32.0	47.7	57.0	64.8	67.2	63.7	55.6	45.3	33.2	24.8	
		MCDB	21.3	22.9	34.2	56.7	70.2	76.0	76.3	74.3	65.5	51.2	35.0	26.1	
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.7	14.3	17.5	20.6	18.7	17.9	19.8	18.5	14.0	11.4	13.3	
		MCDBR	14.5	14.1	13.5	25.3	26.5	23.8	22.3	25.2	27.3	21.9	12.2	13.9	
		MCWBR	14.5	13.9	11.2	14.0	12.0	9.7	8.7	10.2	12.0	12.0	9.7	13.9	
	5% WB	MCDBR	14.4	13.7	12.1	23.8	23.3	20.2	18.6	21.9	23.6	19.7	12.0	13.6	
		MCWBR	14.4	13.7	10.6	14.0	12.4	9.8	8.7	10.3	11.4	12.1	10.1	13.7	
Clear-Sky Solar Irradiance	taub	0.294	0.326	0.380	0.405	0.407	0.410	0.436	0.418	0.371	0.352	0.328	0.297		
	taud	2.021	1.972	1.913	2.138	2.201	2.278	2.200	2.222	2.304	2.236	2.082	1.977		
	Ebn at Noon	196	230	244	259	266	266	256	253	251	224	180	162		
	Edh at Noon	23	35	47	43	42	40	42	39	31	27	22	20		
All-Sky Solar Radiation	RadAvg	331	671	1143	1497	1826	1949	1765	1497	1056	551	263	220		
	RadStd	28	35	89	103	155	151	121	115	101	66	36	29		